R documentation

of all in 'man'

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`baserate` *Calculate Baserate* 

Description

This function will calculate the baserate of the first and second rater as well as the average of both the raters.

Usage

```
baserate(set)
```

Arguments 

`set`	The `set` for which the baserate is calculated

Details

A baserate is the percentage, as a decimal, that a positive code appears in a set for a given rater. It is assumed that the first rater is more experienced and thus provides a better estimation of the actual baserate for a given code. If the raters are assumed to have the same experience level, the average baserate may give a more accurate approximation. If the second rater is more experienced, the second rater's baserate would be assumed to be a better estimation. Functions assume that the first rater is the more experienced rater and thus uses the first rater's baserate as the overall baserate estimation.

Value

A list of baserates; first being the first rater's, second being the second rater's, and the third being the average of the two raters.

`contingencyTable` *contingencyTable*

Description

A contingency Table is a 2x2 matrix that contains the counts of all combinations of positive and negative ratings made by two raters.

The first rater values are represented by the first letter, and the second rater values are represented by the second letter. The baserate is calculated by (PP + PN)/(PP + PN + NP + NN).



sage

```
contingencyTable
```

Format

The contingency Table is an object of class data.frame: as.data.frame(matrix(c(PP, NP, PN, NN))).

Examples

```
#A example contingencyTable
ct = as.data.frame(matrix(c(3,2,1,34), nrow = 2, ncol = 2))

#This contingencyTable is included in the package under the variable name "contingencyTable".
```

createSDS *Create Simulated Data Set*

Description

Creates a simulated data set based off the given parameters

Usage

```
createSDS(length, baserate, kappaMin, kappaMax, precisionMin, precisionMax)
```

Arguments

length	the length of the simulated data set to be created
baserate	the baserate of the simulated data set
kappaMin	the minimum kappa of the simulated data set
kappaMax	the maximum kappa of the simulated data set
precisionMin	the minimum precision of the simulated data set
precisionMax	the maximum precision of the simulated data set

Details

Sets are generated by first picking a random kappa within the range and a random precision within its range. If the random kappa, random precision, and baserate are not mathematically possible, then the precision is resampled from a range of mathematically possible values. A unique simulated data set is then constructed given these parameters.

Value

returns a simulated data set that fulfills the given parameters

---

kappaCT *Calculate kappa (contingency Table)*

Description

This function calculates kappa given a contingency table.

Usage

```
kappaCT(ct)
```

Arguments

ct	A contingencyTable

Value

Kappa as a double.

Examples

```
#Given a contingency Table
kappaCT(contingencyTable)
```

---

kappaSet *Calculate kappa (Set)*

Description



This function calculates kappa given a set.

Usage

```
kappaSet(set)
```

Arguments

set	A set

Value

Kappa as a double.

Examples

```
# Given set
kappaSet(set)
```

---

rho *Calculate rho*

Description

This function will calculate rho given the parameters of an observed test set, which it returns as a double.

Usage

```
rho(observedKappa, SDSBaserate, testSetLength, testSetBaserateInflation = 0,
  SDSLength = 10000, replicates = 800, kappaThreshold = 0.65,
  SDSKappaMin = 0.4, SDSPrecisionMin = 0.6, SDSPrecisionMax = 1)
```

Arguments

observedKappa	The observed kappa value that will be tested with rho, as a double
SDSBaserate	The baserate for a simulated data set
testSetLength	The number of items evaluated by two raters to establish inter rater reliability
testSetBaserateInflation	The percent, as a decimal, of the test set that is guaranteed to be a positive for the first rater
SDSLength	The length of the simulated data sets used to calculate rho
replicates	The number of simulated data sets to use in the null hypothesis distribution for rho
kappaThreshold	The maximum kappa value used to generate simulated data sets in the null hypothesis distribution for rho
SDSKappaMin	The minimum kappa value used to generate simulated data sets in the null hypothesis distribution for rho
SDSPrecisionMin	The minimum precision to be used for generation of simulated data sets in the null hypothesis distribution for rho
SDSPrecisionMax	The maximum precision to be used for generation of simulated data sets in the null hypothesis distribution for rho

Value

rho, as a double

Note

replicates as used in m  carlo studies.

Examples

```
# Given only required parameters
rho(observedKappa = 0.88, SDSBaserate = 0.2, testSetLength = 80)

# Given required parameters plus testSetBaserateInflation and runs
rho(
 observedKappa = 0.88, SDSBaserate = 0.2, testSetLength = 80,
 testSetBaserateInflation = 0.2, replicates = 500
)
```

---

rhoCT *Rho (contingency Table)*

Description

This function calculates rho and kappa for a given contingency table, and returns a list containing both values.

Usage

```
rhoCT(table, SDSBaserate = NULL, testSetBaserateInflation = 0,
  SDSLength = 10000, replicates = 800, kappaThreshold = 0.65,
  SDSKappaMin = 0.4, SDSPrecisionMin = 0.6, SDSPrecisionMax = 1)
```

Arguments

table	The contingencyTable used to calculate rho
SDSBaserate	The baserate of a simulated data set, if NULL, assume the baserate of the simulated data set is the same as the test
testSetBaserateInflation	The percent, as a decimal, of the test set that is guaranteed to be a positive for the first rater
SDSLength	The length of the simulated data sets used to calculate rho
replicates	The number of simulated data sets to use in the null hypothesis distribution for rho
kappaThreshold	The maximum kappa value used to generate simulated data sets in the null hypothesis distribution for rho
SDSKappaMin	The minimum kappa value used to generate simulated data sets in the null hypothesis distribution for rho
SDSPrecisionMin	The minimum precision to be used for generation of simulated data sets in the null hypothesis distribution for rho
SDSPrecisionMax	The maximum precision to be used for generation of simulated data sets in the null hypothesis distribution for rho

Value

list containing rho and kappa

Note

replicates as used in monte carlo studies.

Examples

```
# Given testSetBaserateInflation
rhoCT(contingencyTable, testSetBaserateInflation = 0.2)

# Given SDSBaserate and testSetBaserateInflation
rhoCT(contingencyTable, SDSBaserate = 0.1, testSetBaserateInflation = 0.2)
```

`rhoMinTSLength` *Rho Min testSetLength*

Description

This function calculates the minimustSetLength where it is possible to get a rho less than alpha, which it returns as an integer.

Usage

```
rhoMinTSLength(baserate, alpha = 0.05, inc = 10, printInc = FALSE, ...)
```

Arguments

`baserate`	A `baserate`
`alpha`	A double indicating the threshold of significance for rho (similar to an alpha level for a p value), defaulted to 0.05
`inc`	An integer indicating by how much the testSetLength should increase each iteration
`printInc`	A boolean indicating whether to print out each increment value with it's corresponding significance for rho
`...`	Any additional parameters passed into `rho`

Value

An integer giving the minimum length of testSet, as a denomination of inc, that would give a value where rho becomes mathematically possible

Examples

```
#Add testSetBaserateInflation as an additional parameter
rhoMinTSLength(0.2, testSetBaserateInflation = 0.33)

#Add testSetBaserateInflation as well as changing inc and selecting printInc
rhoMinTSLength(0.2, inc = 5, printInc = TRUE, testSetBaserateInflation = 0.33)
```

`rhoR` *rhoR: A package for computating rho.*

Description

rho is used to test the generalization of inter rater reliability (IRR) statistics, in this case Cohen's Kappa.

Details

Rho starts with an observed kappa value, calculated on a subsample of data, known as an observed test set, and a *kappa threshold* which indicates significant agreement between raters.

It then generates a collection of fully-coded, *simulated data sets* (SDS), further described in `createSDS`, all of which have a kappa value below the kappa threshold.

Then kappa is calculated on a ~~simulated~~ *test set* sampled from each of the SDSs in the collection to create a null hypothesis distribution. These simulated test sets mirror the observed test set in their size and sampling method.

The null hypothesis is that the observed test set~~,~~ was sampled from a SDS within a population of simulated data sets which have kappas less than SDS kappa max, or that the observed kappa is not representative of the level of agreement that would be reached if both raters were to code the entire data set.

Researchers can set their own alpha level. If alpha is greater than the percentage of the distribution of samples from the code sets that is greater than the observed kappa, the null hypothesis is rejected and one can conclude that two raters would have acceptable agreement had they coded the rest of the data.

rho

Use `rhoSet`
`rhoCT`

kappa

Use `kappaSet`
`kappaCT`

rhoMinTSLength

Use `rhoMinTSLength`

`rhoSet` *Rho (set)*

Description

This function calculates rho and kappa for a given set, and returns a list containing both values.

Usage

```
rhoSet(set, SDSBaserate = NULL, testSetBaserateInflation = 0,
  SDSLength = 10000, replicates = 800, kappaThreshold = 0.65,
  SDSKappaMin = 0.4, SDSPrecisionMin = 0.6, SDSPrecisionMax = 1)
```

Arguments

`set`	The test `set` used to calculate rho
`SDSBaserate`	The `baserate` of a simulated data set, if NULL, assume the baserate of the simulated data set is the same as the test
`testSetBaserateInflation`	The percent, as a decimal, of the test set that is guaranteed to be a positive for the first rater
`SDSLength`	The length of the simulated data sets used to calculate rho
`replicates`	The number of simulated data sets to use in the null hypothesis distribution for rho
`kappaThreshold`	The maximum kappa value used to generate simulated data sets in the null hypothesis distribution for rho
`SDSKappaMin`	The minimum kappa value used to generate simulated data sets in the null hypothesis distribution for rho
`SDSPrecisionMin`	The minimum precision to be used for generation of simulated data sets in the null hypothesis distribution for rho
`SDSPrecisionMax`	The maximum precision to be used for generation of simulated data sets in the null hypothesis distribution for rho

Value

A list containing rho and kappa

Note

replicates is used in monte carlo studies.

Examples

```
# Given testSetBaserateInflation
rhoSet(set, testSetBaserateInflation = 0.2)

# Given SDSBaserate and testSetBaserateInflation
rhoSet(set, SDSBaserate = 0.1, testSetBaserateInflation = 0.2)
```

`set` *Set*

Description

A set can be any length of vectors that are bound using `cbind`. The length of each of these vectors becomes the set length. The first column of the set is defaulted as the first rater.

Usage

```
set
```

Format

The set is an object of class data.frame with n rows and two columns.

Examples

```
#An example test set
gold = c(1,1,1,1,rep(0,36))
silver = c(1,1,1,0,1,1,rep(0,34))
exampleSet = cbind(gold,silver)

#This set is included in the package under the variable name "set".
```

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